

|                                            |                                         |
|--------------------------------------------|-----------------------------------------|
| Program : <b>Diploma in Architecture</b>   |                                         |
| Course Code : <b>5182C</b>                 | Course Title: <b>Applied Ergonomics</b> |
| Semester : <b>5</b>                        | Credits: <b>4</b>                       |
| Course Category: <b>Program Elective</b>   |                                         |
| Periods per week: <b>4 (L:4, T:0, P:0)</b> | Periods per semester: <b>60</b>         |

### Course Objectives:

- To familiarize the importance of anthropometry in design of spaces
- To explain design parameters for children, differently able and the elderly.
- To explain ergonomics and its application in improving productivity /efficiency

### Course Prerequisites:

| Topic                                         | Course Code | Course name              | Semester |
|-----------------------------------------------|-------------|--------------------------|----------|
| Design process, anthropometry, parking spaces |             | Architectural drawing-II | 4        |
| Lift, ramp                                    |             | Building services        | 4        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO n | Description                                                                                   | Duration (Hours) | Cognitive Level |
|------|-----------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1  | Discuss ergonomic aspects and Occupational hazards in Work environment                        | 12               | Understanding   |
| CO2  | Familiarize the application of anthropometry in designing spaces for children, elderly people | 15               | Applying        |
| CO3  | Interpret the importance of ergonomics at workplace                                           | 14               | Understanding   |
| CO4  | Explain barrier free design and Inclusive Design for barrier free environment                 | 17               | Applying        |
|      | Series Test                                                                                   | 2                |                 |

**CO – PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     | 3   |     |     |
| CO2             | 3   |     |     |     | 3   |     |     |
| CO3             | 3   |     |     |     | 3   |     |     |
| CO4             | 3   |     |     |     | 3   |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes                                                                                                                                                                                                                          | Description                                                                                          | Duration (Hours) | Cognitive Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                      | <b>Discuss ergonomic aspects and Occupational hazards in Work environment.</b>                       |                  |                 |
| M1.01                                                                                                                                                                                                                                    | Explain anthropometry, ergonomics and Biomechanics                                                   | 4                | Understanding   |
| M1.02                                                                                                                                                                                                                                    | Discuss the health issues associated with various environmental conditions in Work Spaces.           | 4                | Understanding   |
| M1.03                                                                                                                                                                                                                                    | Describe various occupational hazards in work environment.                                           | 4                | Understanding   |
| <b>Contents:</b><br>anthropometry, ergonomics and Biomechanics - Environmental condition including, thermal, illumination, noise and vibration-Health issues - Occupational hazards in work environment, Visual stress, Postural Stress. |                                                                                                      |                  |                 |
| CO2                                                                                                                                                                                                                                      | <b>Familiarize the application of anthropometry in designing spaces for children, elderly people</b> |                  |                 |
| M2.01                                                                                                                                                                                                                                    | Discuss movements and activities of human beings in built environment.                               | 4                | Understanding   |
| M2.02                                                                                                                                                                                                                                    | Illustrate anthropometry for children, elderly people                                                | 6                | Applying        |
| M2.03                                                                                                                                                                                                                                    | Explain design parameters for children, elderly people                                               | 5                | Understanding   |
|                                                                                                                                                                                                                                          | Series Test – I                                                                                      | 1                |                 |
| <b>Contents:</b><br>Introduction to Human Movement - Human being in built environment - importance of anthropometry, anthropometry for children, elderly - Design parameters for children, elderly people                                |                                                                                                      |                  |                 |

|                                                                                                                                                                                                                                                                     |                                                                                       |    |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----|---------------|
| <b>CO3</b>                                                                                                                                                                                                                                                          | <b>Interpret the importance of ergonomics at workplace</b>                            |    |               |
| M3.01                                                                                                                                                                                                                                                               | Discuss importance of anthropometry of people at workplace                            | 4  | Understanding |
| M3.02                                                                                                                                                                                                                                                               | Illustrate anthropometry of people at work place                                      | 4  | Applying      |
| M3.03                                                                                                                                                                                                                                                               | Describe ergonomics at work place                                                     | 6  | Understanding |
| <b>Contents:</b><br>Ergonomics and Design - Introduction to Anthropometrics, static and dynamic anthropometrics - Muscles and work physiology - Static and Dynamic work including maximum capacity, Human Efficiency - ergonomics at work place - design parameters |                                                                                       |    |               |
| <b>CO4</b>                                                                                                                                                                                                                                                          | <b>Explain barrier free design and Inclusive Design for barrier free environment.</b> |    |               |
| M4.01                                                                                                                                                                                                                                                               | Discuss barrier free design, Disability, Ageing and differently abled People.         | 3  | Understanding |
| M4.02                                                                                                                                                                                                                                                               | Identify the limitations related to differently abled and elderly people              | 4  | Understanding |
| M4.03                                                                                                                                                                                                                                                               | Apply criteria for inclusive design in a plan                                         | 10 | Applying      |
|                                                                                                                                                                                                                                                                     | Series Test – II                                                                      | 1  |               |
| <b>Contents:</b><br>Barrier free design - Disability, Ageing and differently able - Ramp, toilets and corridor design - Spatial Requirements for wheel chair movement - Design criteria, Inclusive Design for barrier free environment                              |                                                                                       |    |               |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                                                                 |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1         | Chaira, J. D. and Callender, J. H. (1987). Time Savers Standards for Building Types. Singapore: McGraw-Hill.                                                             |
| R2         | Crosbie, M. J. and Watson, D. (2005). Time Savers Standards for Architectural Design: Technical data for Professional Practice. 8th Ed. The McGraw-Hill Company.         |
| R3         | Handbook on Barrier free and Accessibility by CPWD                                                                                                                       |
| R4         | Design Manual for a Barrier Free Environment in Universities/Colleges for All India Council for Technical Education (AICTE) Presented by SVAYAM FOUNDATION & CBR NETWORK |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | aged&disabled.PDF (cpwd.gov.in)                                                                                                                                         |
| 2            | <a href="https://www.aicte-india.org/downloads/Svayam-AICTE%20Guidelines-modified.pdf">https://www.aicte-india.org/downloads/Svayam-AICTE%20Guidelines-modified.pdf</a> |